



10455 - Confirming surprisingly high metallicities in galaxies in the Epoch of Reionization

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	REBELS-05	NIRSpec IFU Spectroscopy	(1) REBELS-05
	2	REBELS-29	NIRSpec IFU Spectroscopy	(2) REBELS-29
	3	REBELS-32	NIRSpec IFU Spectroscopy	(3) REBELS-32

ABSTRACT

How galaxies build up their mass in the early Universe remains one of the most puzzling questions of modern cosmology, especially since recent discoveries have revealed massive galaxies with significant dust reservoirs during the Epoch of Reionization (EoR). To understand how these galaxies grew so rapidly, we propose deep NIRSpec IFU spectroscopy for three of the most evolved and massive star-forming galaxies at $z \sim 7$, all of which have surprisingly high, but uncertain, metallicity estimates. These massive galaxies are ideal laboratories for probing early galaxy assembly and chemical evolution. Our observations will target key auroral lines ([OIII]4363) and other crucial rest-frame optical emission lines, allowing us to directly measure their electron temperatures and confirm their metallicities. The redshift range of these targets also enables us to explore their electron densities, nitrogen-to-oxygen abundances, ionization parameters, star formation rates, extinction, and other key ionized gas properties with observations in a single NIRSpec filter.

By combining these data with existing ALMA [CII] and dust continuum observations, we will obtain a spatially resolved view of the warm ionized gas, cold neutral gas, metals, and dust within these evolved EoR galaxies, providing critical insights into their assembly. Additionally, this unique combination of JWST and ALMA data is critical to explore the interplay between dust and metals, test early dust production mechanisms, and link these properties to the galaxies' kinematics and turbulence in the interstellar medium, offering a comprehensive understanding of how massive systems evolved in the early Universe.

OBSERVING DESCRIPTION

We will conduct NIRSpec IFU G395M/F290LP spectroscopy for three galaxies: REBELS-05 ($z = 6.496$), REBELS-29 ($z = 6.685$), and REBEL-32 ($z = 6.729$).

We use the NRSIRS2 readout mode to avoid data volume issues with our observations. To ensure detections of [OIII]4363, we require 15 ks exposures for REBELS-05, 14 ks for REBELS-29, and 21 ks for REBELS-32. Each individual integration is set to be less than the maximum recommended number of groups (<20 for NRSIRS2).

We use a 4-point dither pattern for extended sources to reduce the impact of contamination of the spectra from failed microshutters. In addition, with this dithering strategy we do not require leak-cal. Given that we are interested in emission lines only we do not design dedicated background observations, but will use empty pixels in the galaxy-free regions of the IFU FoV for background subtraction.

We do not require a specific target acquisition image as an error of the order of the JWST absolute accuracy (0.1 arcsec) would not effect our observations.

Proposal 10455 - Targets - Confirming surprisingly high metallicities in galaxies in the Epoch of Reionization

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	REBELS-05	RA: 02 18 11.5100 (34.5479583d) Dec: -05 00 59.30 (-5.01647d) Equinox: J2000		
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies] Extended=YES				
	(2)	REBELS-29	RA: 10 01 36.8500 (150.4035417d) Dec: +02 37 49.10 (2.63031d) Equinox: J2000		
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies] Extended=YES				
	(3)	REBELS-32	RA: 10 01 59.0700 (150.4961250d) Dec: +01 53 27.50 (1.89097d) Equinox: J2000		
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies] Extended=YES				

Proposal 10455 - Observation 1 - Confirming surprisingly high metallicities in galaxies in the Epoch of Reionization

Observation	Proposal 10455, Observation 1: REBELS-05											Mon May 11 19:00:42 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	REBELS-05	RA: 02 18 11.5100 (34.5479583d) Dec: -05 00 59.30 (-5.01647d) Equinox: J2000									
	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395M/F290LP	NRSIRS2	17	3	false	true	NONE	4	12	15055.734	
Special Requirements	Aperture PA Range 226.97164917 to 53.97164917 Degrees (V3 88.0 to 275.0)											

Proposal 10455 - Observation 2 - Confirming surprisingly high metallicities in galaxies in the Epoch of Reionization

Observation	Proposal 10455, Observation 2: REBELS-29											Mon May 11 19:00:42 GMT 2026
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 2:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(2)	REBELS-29	RA: 10 01 36.8500 (150.4035417d) Dec: +02 37 49.10 (2.63031d) Equinox: J2000									
Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies] Extended=YES												
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395M/F290LP	NRSIRS2	16	3	false	true	NONE	4	12	14180.401	
Special Requirements	Aperture PA Range 13.97164917 to 91.97164917 Degrees (V3 235.0 to 313.0)											

Proposal 10455 - Observation 3 - Confirming surprisingly high metallicities in galaxies in the Epoch of Reionization

Observation	Proposal 10455, Observation 3: REBELS-32 Diagnostic Status: Warning Observing Template: NIRSspec IFU Spectroscopy											Mon May 11 19:00:42 GMT 2026
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	REBELS-32	RA: 10 01 59.0700 (150.4961250d) Dec: +01 53 27.50 (1.89097d) Equinox: J2000									
Template	Comments: Category=Galaxy Description=[Emission line galaxies, High-redshift galaxies] Extended=YES											
	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395M/F290LP	NRSIRS2	18	4	false	true	NONE	4	16	21241.424	
Special Requirements	Aperture PA Range 244.97164917 to 162.97164917 Degrees (V3 106.0 to 24.0)											